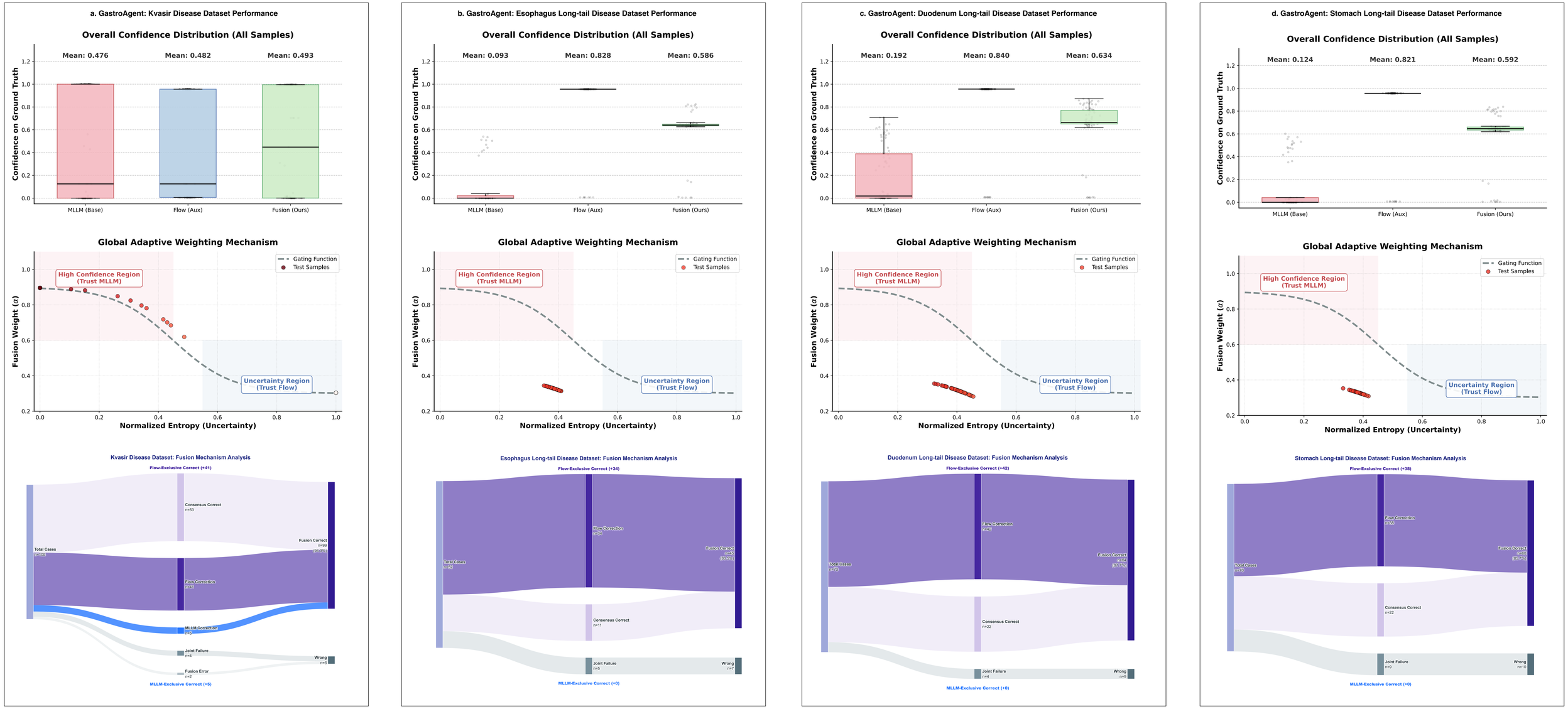
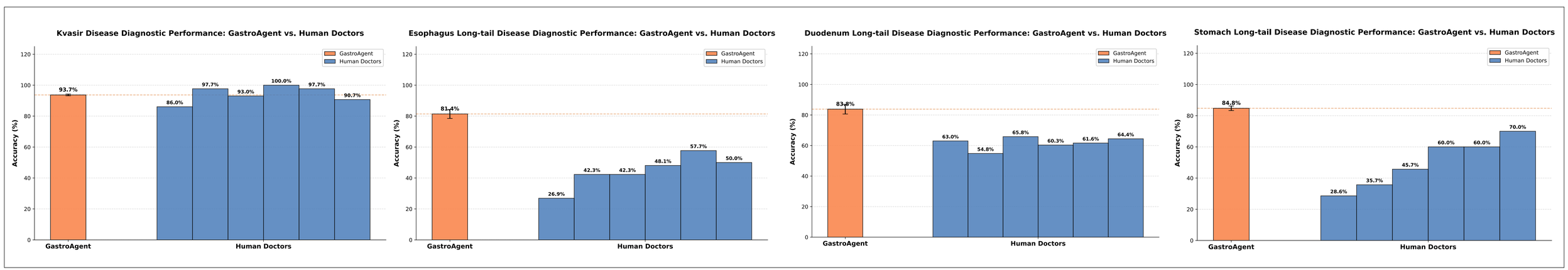

a. GastroAgent: Kvasir Disease Dataset Performance
Overall Confidence Distribution (All Samples)
Mean: 0.476
Mean: 0.482
Mean: 0.493
Confidence on Ground Truth
MLLM (Base)
Flow (Aux)
Fusion (Ours)
Global Adaptive Weighting Mechanism
Gating Function
Test Samples
High Confidence Region (Trust MLLM)
Uncertainty Region (Trust Flow)
Fusion Weight (α)
Normalized Entropy (Uncertainty)
Kvasir Disease Dataset: Fusion Mechanism Analysis
Flow-Exclusive Correct (+41)
Total Cases n=100
Consensus Correct n=53
Flow Correction n=41
MLLM Correction n=5
Joint Failure n=4
Fusion Error n=2
Fusion Correct n=99 (94.3%)
Wrong n=6
MLLM-Exclusive Correct (+5)
b. GastroAgent: Esophagus Long-tail Disease Dataset Performance
Overall Confidence Distribution (All Samples)
Mean: 0.093
Mean: 0.828
Mean: 0.586
Confidence on Ground Truth
MLLM (Base)
Flow (Aux)
Fusion (Ours)
Global Adaptive Weighting Mechanism
Gating Function
Test Samples
High Confidence Region (Trust MLLM)
Uncertainty Region (Trust Flow)
Fusion Weight (α)
Normalized Entropy (Uncertainty)
Esophagus Long-tail Disease Dataset: Fusion Mechanism Analysis
Flow-Exclusive Correct (+34)
Total Cases n=62
Flow Correction n=34
Consensus Correct n=11
Joint Failure n=5
Fusion Correct n=56 (90.3%)
Wrong n=7
MLLM-Exclusive Correct (+0)
c. GastroAgent: Duodenum Long-tail Disease Dataset Performance
Overall Confidence Distribution (All Samples)
Mean: 0.192
Mean: 0.840
Mean: 0.634
Confidence on Ground Truth
MLLM (Base)
Flow (Aux)
Fusion (Ours)
Global Adaptive Weighting Mechanism
Gating Function
Test Samples
High Confidence Region (Trust MLLM)
Uncertainty Region (Trust Flow)
Fusion Weight (α)
Normalized Entropy (Uncertainty)
Duodenum Long-tail Disease Dataset: Fusion Mechanism Analysis
Flow-Exclusive Correct (+42)
Total Cases n=75
Flow Correction n=42
Consensus Correct n=22
Joint Failure n=4
Fusion Correct n=64 (87.7%)
Wrong n=5
MLLM-Exclusive Correct (+0)
d. GastroAgent: Stomach Long-tail Disease Dataset Performance
Overall Confidence Distribution (All Samples)
Mean: 0.124
Mean: 0.821
Mean: 0.592
Confidence on Ground Truth
MLLM (Base)
Flow (Aux)
Fusion (Ours)
Global Adaptive Weighting Mechanism
Gating Function
Test Samples
High Confidence Region (Trust MLLM)
Uncertainty Region (Trust Flow)
Fusion Weight (α)
Normalized Entropy (Uncertainty)
Stomach Long-tail Disease Dataset: Fusion Mechanism Analysis
Flow-Exclusive Correct (+38)
Total Cases n=70
Flow Correction n=45
Consensus Correct n=22
Joint Failure n=9
Fusion Correct n=62 (88.6%)
Wrong n=10
MLLM-Exclusive Correct (+0)

e.


Kvasir Disease Diagnostic Performance: GastroAgent vs. Human Doctors
GastroAgent
Human Doctors
Accuracy (%)
93.7%
86.0%
97.7%
93.0%
100.0%
97.7%
90.7%
Esophagus Long-tail Disease Diagnostic Performance: GastroAgent vs. Human Doctors
81.4%
26.9%
42.3%
42.3%
48.1%
57.7%
50.0%
Duodenum Long-tail Disease Diagnostic Performance: GastroAgent vs. Human Doctors
83.8%
63.0%
54.8%
65.8%
60.3%
61.6%
64.4%
Stomach Long-tail Disease Diagnostic Performance: GastroAgent vs. Human Doctors
84.8%
28.6%
35.7%
45.7%
60.0%
60.0%
70.0%